

COMBINED SOURCES SOUGHT NOTICE AND NOTICE OF INTENT TO SOLE SOURCE

Announcement Type: Special Notice

Announcement Number: AMDTCNOI2601351

Subject: Dilution Refrigerator Magnet for Quantum Memory

Applicable NAICS: 334516 — Analytical Laboratory Instrument Manufacturing with a small business size standard of 1,000 employees.

Applicable PSC: 6640 – Laboratory Equipment and Supplies

Purpose

This is a combined sources sought notice and notice of intent to sole source, not a request for a quotation. NIST is seeking information from sources that may be capable of providing a solution that will achieve the essential requirements included in the “Description” section below.

If no alternate sources are identified, the Government intends to negotiate a firm-fixed price purchase order on a sole source basis with Bluefors, Inc., located at 253 36th Street Suite C663, Brooklyn, New York 11232-2546 USA. This action will be conducted under the authority of FAR RFO Part 12.102(a) and 41 USC 1901, allowing the Contracting Officer (CO) to solicit from one source. Delivery will be required within eight (8) months after contractor receipt of order.

Description

The Contractor shall deliver a quantity of one (1) 3D Vector Magnet with Power Supply and Programmer (fully compatible with Bluefors LD 250 Dilution Refrigerator), inclusive of FOB Destination delivery and shall meet the minimum requirements identified below.

- a. Rated Field(Z-Y-X): 6-1-1 Tesla at 4.2K
- b. Homogeneity(Z-Y-X): +/-0.5%(Z), +/-1%(Y), +/-1.5%(X)
- c. Persistent Current Switch to enable charging superconducting magnets without continuous power supply.
- d. Three axis Power Supply and Programmer to drive and manage the 3D vector superconducting magnet system

Background

In collaboration with the University of Chicago and Carnegie Mellon University, the National Institute of Standards and Technology (NIST) Information Technology Laboratory (ITL) is developing a silicon carbide (SiC) microcavity-enhanced quantum memory based on erbium-doped yttrium orthosilicate (Er:YSO). Quantum memory is a critical component for quantum communication and networking systems, particularly in enabling quantum repeaters and distributed quantum network nodes. The SiC microcavity-enhanced Er:YSO platform represents a promising approach for achieving high-efficiency quantum memory at telecommunications wavelengths. As part of the Quantum Communication and Networking (QuCaN) project within

NIST ITL, this effort aims to advance the performance and integration of quantum memory technologies in practical network environments.

The NIST laboratory currently operates a Bluefors LD250 dilution refrigerator, which provides a base temperature of approximately 10 mK. However, the system is not equipped with a magnet. To enable the required experimental capabilities, it is necessary to procure a 3D vector magnet system that is fully compatible with the existing dilution refrigerator.

Although multiple vendors offer magnet systems designed for cryogenic environments, Bluefors is the only known source capable of providing a magnet system that meets the specific integration requirements of our existing LD250 platform. The required magnet must be mechanically, electrically, and cryogenically compatible with the current system. Only Bluefors manufactures magnet systems that are specifically designed, factory-certified, and guaranteed to integrate seamlessly with the LD250 dilution refrigerator.

How to Respond to this Notice

Interested parties that believe they could satisfy the requirements listed above for NIST may clearly and unambiguously identify their capability to do so in writing by or before the response date for this notice. This combined sources sought notice and notice of intent to sole source is not a solicitation.

All responses to this notice must be submitted via email to Jennifer Lohmeier via email at Jennifer.lohmeier@nist.gov so that they are received no later than **May 8, 2026 at 10:00 AM** Eastern Time.

Each response should include the following Business Information:

- a. Contractor Name, Address,
- b. Point of Contact Name, Phone Number, and Email address
- c. Contractor UEI
- d. Contractor Business Classification (i.e., small business, 8(a), woman owned, HUBZone, veteran owned, etc.) as validated in System for Award Management (SAM). All offerors must have an active registration in www.SAM.gov.
- e. Capability Statement describing what your company is capable of providing to meet the Government's needs

Important Notes

The information received in response to this notice will be reviewed and considered so that the NIST may appropriately solicit for its requirements in the near future. This notice should not be construed as a commitment by the NIST to issue a solicitation or ultimately award a contract. This notice is not a request for a quotation. Responses will not be considered as proposals or quotations. No award will be made as a result of this notice. NIST is not responsible for any costs incurred by the respondents to this notice. NIST reserves the right to use information provided by respondents for any purpose deemed necessary and appropriate.